Unit 40 - Computer Game Design
Unit 23 - Human Computer Interaction
Assignment 1 - Game Review

Introduction

Scenario / Vocational Context - *You are an indie game maker hoping to get your first game idea to market. Before you start designing your own game, you need to carry out research into some of the key principles of game design; which include the elements of visual style and gameplay.*

Task 1

(U40:P1,M1,D1) - Describe, Explain and Critically evaluate visual style and elements of gameplay

Introduction



In this review, I will be talking about Metro 2033 Redux, a remastered version of the Metro 2033, with it's visual style and gameplay, describing, explaining and critically evaluating these aspects of the game. Since the Redux edition is a graphical remaster of the game, the games are almost identical apart from visuals, so all the gameplay and narrative aspects still apply.

Metro 2033 is a survival horror first-person shooter video game with a PEGI rating of 16, based on the novel with the same name "Metro 2033", by Russian author Dmitry Glukhovsky. It was

developed by 4A Games in Ukraine, and was released on Microsoft Windows and Xbox 360 in March 2010.

The next sequel adding to the Metro franchise, is Metro: Last Light, which was released in May 2013, and the upcoming Metro game, Metro Exodus, has an initial release date of 2018 and it is being developed by the same company on their significantly improved 4A Engine, which is apparently going to be set in an open world map, instead of the usual linear aspect of Metro.

Visual Style

The Game World and Environment

It is widely believed that Metro 2033 has some of the most atmospheric environments ever seen in a video game, making most the players praise the 4a engine due to it's ability to make the game gorgeous to look at, which I personally agree with.

With that being said, the game has two key different game worlds and environments, both of them being the tunnel system of the metro and the surface world. They are vastly different visually and environmentally from each other but both have their dangers to the player. Below are screenshots of the different environments the player may come across in Metro 2033.





With first impressions, the player can easy tell that the surface world is an empty wasteland which was completely wiped out by nuclear war, and that the metro system has a dark and stingy environment due to the nature of an underground tunnel system which was poorly maintained.

I believe that the visuals like this also adds to the story, creating environment that looks wartorn and poorly kept together gives a feeling of suffering and desperation to the surviving populus. Giving the illusion that the best that can be obtained isn't the best, but it is better than nothing.

This type of blending with the terrain used for each level has had attention to detail by the developers, which is usually lacking with most first person shooters and similar games.

Also, with the terrain, it is noticeable that the developers have done their research to make sure that the buildings and other structures, such as non-movable and non-destructible objects fits and blends in well with the entire terrain.

Having said that, since most the architecture in metro is non-destructible, there are some scripted events within the game where it shows the architecture, such as a previously non-destructible car or a large wall on a building being destroyed, to make the illusion that the environment can be interacted with, but that the player doesn't have the strength to do so.

There are also some objects which can be interacted with, ranging from turning lights on and off to picking up ammo from enemy weapons and even the weapon itself. The objects in the game has a subtle but noticeable pattern, making the player notice it is intractable, but they also blend

in with the rest of the environment, so they don't stand out completely. In my opinion, having interactable objects is extremely important to this game due to the survival nature of the gameplay.

However, a big downside I have to this is that the graphics on Metro 2033 are aging significantly due to the game being considered old, even for the remastered version from 2014, even though the graphics are significantly better in the redux edition.

This is a massive downside to Metro, the reason being that it is immersive due to its atmosphere, and I feel that it is slowly losing it with the Redux edition and that it has already lost it on the original version from 2010, due to the improvement of graphics in recent games making the older games look worse.

Characters

Since Metro 2033 is a game which is strongly focused on the story, due to the fact it is based on a novel, the characters within the game are significantly important for the structure of the game, hence why Metro 2033 has a large and diverse set of characters. Only a few are notable within the game, but they are all well written.

Especially with the main protagonist to the franchise, which is the player controlled character Artyom. This character is the only player controlled character in the entire game, making the rest non-player characters, or NPCs. The reason for this is due to the story being written in a first person view from a single survivor.

Since Artyom is the only player controlled character, it doesn't mean that this is the only character you will know the story to. The rest of the characters within the game have their own unique features to them visually and to their personality, so there is hardly a feeling of repetition when interacting with other characters.

The AI system within the game to create the friendly and enemy AI isn't very special when compared to other games while carrying out generic objectives, such as moving around or are busy in combat. However, the AI is very good at communicating with each other, making it feel to the player that they are real people with full personalities talking to each other.

One feature that is noticeable is that the characters are created visually to their personality. What I mean by this is that, the characters that are found to have military equipment or weaponry tend to act more aggressive or alert to their surroundings, while others that have equipment purely to survive or having a lack or no military equipment would be too desperate to interact with or would ask the player for help.

I believe the character designs are well made and feel real within the rest of the world on Metro. The reason I think this is due to the characters that the player will come across will either look

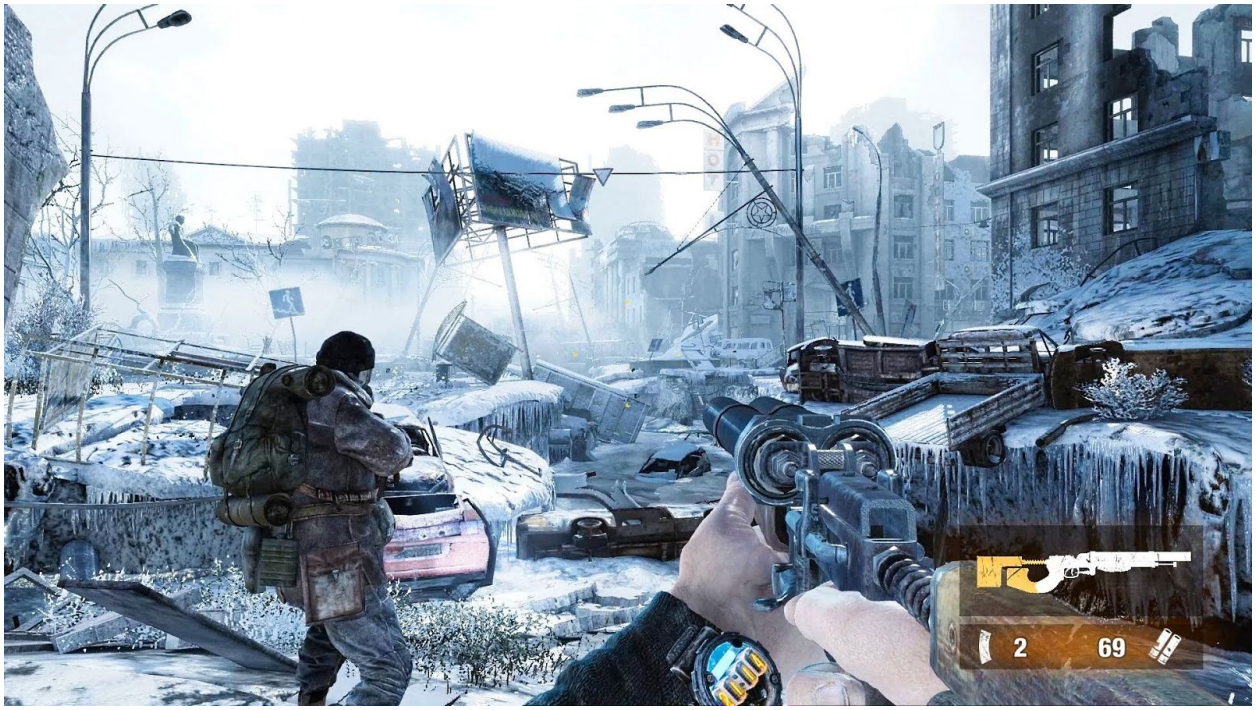
like they are trying desperately to survive, due to distinctive features such as the visual and audial symptoms of illness and malnutrition, or that they are trying to find the best equipment to fight against the threat which is found throughout the city.

The downside I have for the character design is both the same reason for the environment and that the AI are becoming less impressive, due to the improvement of other games. The graphics making the characters look less immersive due to the game aging when compared to newer ones.

Also with the previously thought intelligent AI of the game is also losing its immersion due to games with better AI being released. Making the characters of the game, which is important for a game like Metro, seem less like humans and making the scripted AI stand out.

Interface

The interface in Metro 2033 I find to be extremely minimalistic, which has been done deliberately to make the environment stand out and the game more immersive. The UI only shows the essentials for the current scenario, such as reloading a weapon or choosing different equipment.



However, a well made aspect of Metro which I have noticed very quickly is the ability for the player to recognise what to do without the UI telling the player that there is an action to be carried out.

For example, while playing the game on the surface world, due to the unbreathable air and radioactivity, I was forced to wear a gas mask. The gas mask has an air filter which needs to be swapped out depending on the quality of the air filter, which more can be found through scavenging.

When the air filter of the gas mask is close running out, the UI doesn't give you a warning, unless the player opens up the inventory screen. Instead, the player will notice Artyom breathing more heavily in an unnatural manner, due to Artyom not breathing enough clean air, therefore meaning that the filter needs to be swapped.

Another example of this is when the gas mask visor gets dirty, through dirt or blood landing on the visor. Which the player can clean off by tapping the designated key, making it easier to see. Not only get it get dirty however, even the gasmask visor breaking through extensive damage taken from hostile NPCs.

Which the player will notice through the cracks on the screen and the heavy breathing through Artyom, meaning that the gasmask needs to be swapped with a new one. Which can be found through scavenging but cannot be stored in the inventory, giving a feeling of tension due to the emergency.

Even while having virtually no UI while playing the game, the player can still see the ammo on the weapon, ammo in reserve and the amount of military grade bullets (These type of bullets are used as both currency for shops or as a bullet with higher damage) the player has.



As you can see in the image above, there is also a very detailed UI if the player presses the designated button to open up the inventory, giving the player the ability to choose the desired weapon or equipment needed for the situation.

The downside I have to having the UI like this is that it might not give enough information for other type of players, sometimes including myself. One significant problem I had a few times which caused an annoyance was the lack of a minimap or any method to show which way the player needs to go, especially of the surface maps.

Since the environment has been made to blend in well, making it feel like a realistic setting, means that nothing stands out very well. Which makes navigating along a little bit tedious sometimes, especially when the player isn't accompanied with an NPC to follow.

Perspective

Metro 2033 has a three dimensional perspective with a first person view, since it fits the way the story of the game and the novel it is based off is written. This also adds to the immersion that the game wants to achieve with the player.

I believe that this is probably the best perspective for a game like this, due to its ability to make the player feel like they are Artyom seeing a war torn world with his own eyes, witnessing the disasters that has happened and is happening.

If it was a third person view, I reckon that it is possible to be done. I personally think that it wouldn't feel as rich and genuine as the first person view, but this is a mostly subjective aspect of a game, instead of fact.

However, I genuinely believe that areal view or a 2D perspective would reduce the immersion of the game completely, especially since the story has a protagonist written in a first person perspective.

The downside I have with the first person view, is the fact that it could be a little too immersive for players, which can cause some players to experience too much stress and anxiety due to the nature of the game, making them not want to play the game in general. Also, the opposite can happen with the first person immersion, making it more addictive to people playing the game, making it an unhealthy obsession.

Full Motion Video

Metro 2033 doesn't have the implementation of full motion videos, but it does feature cutscenes which are rendered in-game. These cutscenes are usually played when a significant part of the story is being carried out or that there are scripted events happening that wouldn't be possible to do outside of a cutscene, due to the nature of the gameplay.

I reckon this is a better method because it doesn't break the immersion by showing a video with better graphics even when compared to maximum settings, making the game look worse, or even the 25-30 fps rendering of the video, opposed to the above 60fps the game runs.

I believe that the cutscenes in the game are very wellmade and that it doesn't break the immersion. Instead, adding to the immersion due to the scripted animations looking more natural to what a human would do, which aren't possible to do with a keyboard or controller.

The downside I have with cutscenes and FMVs in general is that it could make the player frustrated, especially if they get impatient easily. There are different people, who experience cutscenes, that either want to skip it instantly or want to find out more about the story through a cutscene.

Giving the ability to skip a cutscene, especially one that is significant to the story, might make the impatient player not know as much as the story or vital information, making the player wander blindly or miss and important fact.

I reckon if the cutscenes were not skippable, the easily impatient player would be annoyed at the fact it cannot be skipped, making the player not concentrate on the story or information during the cutscene very well or just make the player burnt out on the game, quitting it in boredom.

Gameplay

Interaction Model

The interaction model with Metro 2033 has standard features found in a first person shooter, such as the ability to handle and swap different weapons, fire the weapons, reload the weapons and use equipment such as grenades and throwing knives.

Although, there are some interactions which I feel stand out a lot from the game. Such interactions I have already explained in the interface section, which is how the gasmask is implemented within the UI, giving it significant interaction due to the importance of a gasmask in a game like this.

Having the immersion of a working gasmask is what makes this game such a rich gameplay experience when compared to other first person shooters.

The ability to swap out filters due to the bad air on the surface, or any bad air leaked into the tunnels, and that the gasmask can be severely broken through extensive damage received from combat with mutants or other humans, makes me feel the intensity of the responsibility for survival on these type of situations.

Another form of interaction is the ability to purchase and customise weapons in the shops, using the ingame currency, which is military grade ammunition, to purchase weapons and modifications, for better weapons and weapon interaction.

Also, the scavenging aspect of the game I really enjoy. The reason being is that supplies are very scarce, and seeing a deceased human corpse or a crate is a good sight due to the ability to interact with it, revealing supplies such as ammunition, air filters, military grade ammo and sometimes even a new gasmask.

The downside I have to the interaction with metro is that the controls are not very customizable, especially when compared to highly customisable games, such as games run with the fantastic source engine.

The problem I found with the interaction is the mouse sensitivity and that the controls aren't that great by default. Even though the keyboard controls can be remapped to my desire, the mouse sensitivity isn't as in depth, since it is on a fixed slider, which means I am forced to use what they have to offer.

I found this problematic due to the fact I play with very low mouse sensitivity when compared to the average player. The reason being is that I prefer to have precision over mobility and because I have a large mousepad for mobility. However, with metro the lowest sensitivity was still a bit high for me, which I found annoying that I cannot change it to a sensitivity which would be perfect, especially when compared to source engine games.

Singleplayer and multiplayer

Since Metro 2033 is a story based game, it is clear that it has a single player mode within the game. Unfortunately, it is the only gamemode in Metro, it doesn't have a multiplayer mode or even a coop mode to play with. It makes perfect sense for Metro 2033 to be a singleplayer only game, since it is a story heavy game based of a novel.

Since Metro is single player only, it means that the developers have focused on the NPCs a lot more, since NPCs are the only virtually alive entity within the game. Meaning that the story of the game can be more immersive with these intelligently made AI and their integrated illusions of personalities.

I feel that Metro shouldn't of been a singleplayer only game, but not have a generic multiplayer PVP mode. I believe that a co op mode with a friend would of worked fine, but it might sacrifice the protagonist role of Artyom. On that premise, I reckon that the other player/s in co op could play as other significant characters within the Metro story.

I also believe that Metro could pull off a PVP mode on multiplayer, but it would be difficult. Having gamemodes found in popular multiplayer first person shooters such as a team deathmatch or a capture the flag gamemode wouldn't fit well with Metro.

However, having unique game modes similar to a gamemode found in Valve's popular first-person shooter survival horror game, Left 4 Dead 2, Versus game mode, which features a team playing as survivors and another team playing as the infected.

Narrative

Based off the novel with the same name, Metro 2033 is a beautifully gripping, dark and sometimes depressing tale in an alternative future where a nuclear war has decimated the entire planet, leaving survivors in are forced to live throughout metro stations across the country.

To put the story of the game short and simple, most of the game takes place within the Metro system As I have mentioned before, throughout the story, the player controls Artyom as he moves through the ruins of post-nuclear-apocalyptic Russia, mostly through its metro system, although Artyom does venture above ground on rare occasions.

The reason that he is going through the metro system, is that he has a goal to reach the "Polar Station" to seek help, since the station he was stationed in was attacked by entities known as "The Dark Ones", which are considered to be the next step of evolution to the human race, making the human species lose the survival of the fittest.

However, they aren't the only threat. Throughout the journey, there are many more dangerous encounters such as dealing with mutants, neo-nazis, communists, bizarre anomalies and all manner of other sinister and eerie environments, both above and below ground.

In my opinion, I believe the Metro 2033 story is probably the best story I have seen in a video game so far. I found the story to be very gripping, believable and extremely immersive, which I believe is the sole reason I kept playing the Metro franchise. I found it especially immersive when I changed the voices of the characters to the Russian language with english subtitles for translation, making the game setting more believable.

I also found that the story likes to mess with my emotions, by using hallucination of the past city life of Moscow, and sometimes showing hallucinations questioning the player's moral decisions in the game, even if the player didn't notice he was doing something immoral.

I haven't really got any downsides to the story itself since I haven't found any loopholes with the story itself, which is why I enjoyed the story so much. A lot of effort and passion went into the writing, thanks to the author of the Metro 2033 novel.

That being said, there are other downsides I have to some aspects on the story, not the quality of it. Those aspects being the implementation of strong political views which can be seen as controversial.

I feel like implementing the political views of communism and nazisim wasn't really needed for a story such as Metro 2033, but it also doesn't mean that it removes the immersion of it, because it is possible for small political governments to form in a post apocalyptic world, but I just feel like that implementing it in the story wasn't really needed

Game setting

In basic terms, Metro 2033 is set in a post nuclear war in Moscow, Russia. Which I feel is a very good setting for a post nuclear war due to the fact Russia is a country which is believable for this type of scenario to happen in, especially Moscow.

The style of the game also fits the game setting very well, while playing the game I always believed that I was in a post war world set in a Russian city, with the characters and the way they communicate and with the way the equipment and weaponry is made in a Russian style.

The temporal setting of Metro 2033 is that it is set in the year 2033, which is, according to the story, 20 years after the devastation of the nuclear war. Obviously, the setting of Metro is fictional due to how the setting was made in the story, but that doesn't mean the game cannot be factually right. Infact, I sometimes found the way the game interacts with the player makes the temporal setting much more believable, due to the story being written with such factual accuracy and research on to the aspects of survival, war and desperation.

Overall, I feel that the temporal setting of Metro 2033 fits the story and genre of a post nuclear war survival horror game extremely well, making it believable and immersive to the player, which I found very common while playing Metro.

The environmental setting of Metro, such as the weather and damage to the game world, I found to be very aesthetically pleasing, especially during cutscenes. However, there are some environmental settings of the game can have effects on the player too, some of which aren't even visible to the player.

Since there are two key different settings in Metro, the tunnels of the metro system and the surface, there are different environmental settings on each of them, both having their own dangers to the player.

For instance, in the surface world, there are environmental settings such as: weather, the time of day, radioactivity and the rubble of the city which can sometimes be destructible if the player is careless.

The settings such as weather, time of day and radioactivity are dangerous to the player, radioactivity being the most obvious. The time and day and weather can make the player more vulnerable to attack, especially at night time or the harsh winter storms of Russia, which reduces visibility significantly, since the player is forced to wear a gasmask.

However, this doesn't make the tunnels of the metro system much safer than the surface, maybe on the stations, but that is because they are garrisoned with military grade security. The tunnels usually feature badly lit areas, unbreathable air or gas and other harmful entities linked with the story.

If the player is on the metro system away from a station, there might be dangers such as the lack of visibility due to darkness, more claustrophobic environments and hostile humans and creatures lurking in the tunnels.

Overall, the environmental setting of Metro 2033 makes the game much more intense, difficult and immersive. These type of environmental aspects and dangers is what makes the player really feel that he is experiencing the difficulties of surviving in a war torn city, where the surface alone is uninhabitable and dangerous to go on.

The emotional setting is very heavy with Metro 2033, due to how the writing of the story focuses on how they make the player feel, which is what makes Metro the game it is. Since the game wants the player to feel the desperation and struggle of survival in a world that is torn apart by war, radiation, lack of food and supplies and inhabited with many forms of hostile life.

While playing Metro, I usually found myself in situations where I was feeling sadness, anxiety and especially fear. Due to the game having a dark and depressing story, which makes you feel sorry for the innocent survivors and people trying to protect them, even when you realise that it is only a game.

I felt the feeling of anxiety during cutscenes featuring hostile creatures and humans, but there are also others parts of the game where the emotional setting is completely unexpected, which causes a quick burst of fear or sadness due to the instant jump in emotional setting.

However, the emotional settings in Metro sometimes gives a feeling of relief or satisfaction, which I usually felt after completing a mission or barely surviving a specific part of the game. Also, I found that there are strong feelings of satisfaction when finishing a combat or stealth mission due to the difficulty I played the game on.

Sometimes the opposite can happen though, such as feelings of frustration or annoyance, which I feel whenever I fail, get killed in a combat mission with either humans or other creatures and especially during stealth missions, which I will be explaining later on.

Overall, I found the environmental setting of metro is very well made, giving me a various amount of feelings of anxiety, fear, tension, satisfaction and frustration. In my opinion, Metro 2033 is probably the most emotionally manipulate game I have ever played, which is why I believe the environmental setting is another near perfect aspect of Metro.

The ethical setting of Metro 2033 seems very vague for me. I tend to believe that this type of genre is more of a subjective thing, instead of being targeted at a specific audience or demographic. So, I reckon Metro 2033 can be enjoyed by anyone, as long as they enjoy story based, linear, survival horror games and enjoy difficult games to a degree, but there is an easier gamemode to play if the player decides to play away from the recommended difficulty.

That being said, I believe that Metro is very well made for people who enjoy those type of games. It's this reason why the game is widely seen as a near perfect game or the best horror survival game by those who have played it, which I mostly disagree with because there are also other games which are also very good at doing this, but Metro also does it well, just not perfectly.

Overall, the ethical setting of metro isn't especially targeted at an audience or demographic, the game was just made with passion for the previously written novel "Metro 2033", and most the people who play metro notice that the game wasn't specifically made for an audience, instead made to tell a story to those who do play it.

Goals, challenges and rewards

The overall goal of Metro is to go find help tracking the entities known as the "Dark Ones", which was the cause of the attack on the northern station known as "Exhibition", which the game has two different endings to depending on the player's actions on the game, but this will be mentioned later.

On the prologue of the game, there is a scene where Artyom is fighting off creatures known as "demons" with some of Artyom's friends, this is the game's attempt at a basic tutorial, an introduction to the type of game it is and the first reason for the main goal.

With the aspect of the basic tutorial, it tells the player basic controls, such as how to use the weapon and movement. However, throughout the game, it gives the player small tutorials depending on the new equipment the player found, instead of having a long tutorial at the beginning of the game explaining everything.

In collaboration with the main goal of Metro, there are smaller goals in the game which either help with the main goal or help other survivour struggling with their survival. However, to help survivors is an optional goal, meaning that this will have an effect later on the game.

Overall, the goal of metro is a simple one, which is to find out about the “Dark Ones” and to figure out what they do and what to do with them when reaching the end of the game. The smaller goals of the game either links to the main goal or to the ethical aspect of the player’s decisions.

The challenges of the game is the multiple stations and the visit to the stations, which can be done by travelling the tunnels or the surface. This is very challenging to the player, especially when playing the recommended gamemode and difficulty, which is the survival gamemode on hardcore difficulty.

Especially with the environmental and AI controlled dangers to the player. Which is what makes the game feel more alive to the player, which is what makes the game challenging and immersive to the player.

Obviously, the goals and challenges of the game are linked to the narrative of the game, due to the goal being caused by an important part of the story, which is artyom finding out about the allegedly superior “Dark Ones”, which gets significantly more apparent near the end of the game.

Overall, the challenges of the game is what makes the backbone to the entire gameplay and what makes it so challenging and immersive, but the challenging aspect can be annoying for some players.

The travelling between each station through hostile terrain, sometimes to even reach hostile station which is sometimes a lot more challenging, due to the overwhelming amount of hostiles, which can sometimes make people stop playing the game due to the frustrating difficulty.

The rewards of the game I found to be very satisfactory, which can sometimes feed to addiction. The rewards can include better supplies such as ammunition, weaponry and other equipment which is useful or they can be the completion of a mission, which makes the player progress and find out more about the story and game.

However, the game doesn’t feature many rewards, only the ones that I have mentioned. This can be disadvantageous due to the fact people may see the game as too much effort for very little reward, which tends to be understandable due to the intense difficulty the game can offer.

So, overall, the rewards can be either very satisfactory or not enough rewards for the effort put in, it just depends what type of player is playing the game and how they enjoy playing games in general. Which is why some people may not like playing metro, but enjoy the story of metro.

Player Actions

In Metro 2033, the player's actions has a large impact on the game, especially during the end parts of the game. The players actions are determined throughout the entire time they have played the game, depending on the moral or immoral actions that player has done throughout the entire game.

Basically, the player actions on Metro determines which ending the player will get at the end, even if it is the ending they didn't expect or wanted. Meaning, when they reach the ending, one of two endings will be portrayed, depending on the moral choices the player has made during the game.

The two endings reflect the moral choices of the player, which can be considered as an emotional setting of the game, which I believe is intentional, making the player feel guilty or morally correct because of their actions, or vice versa.

The immoral ending, or "The canonical ending", is when the player, as Artyom, lets the "dark ones" be destroyed by missiles. Which is stationed at a missile silo known as "D6" which Artyom got told, by his friend known as Miller, had enough firepower to destroy the dark ones.

However, this was seen as a backup plan but since most people believed they were hostile, they all agreed in this plan. Then, the game again decides to manipulate the player's emotion and perspective by showing a hallucination which is induced by a Dark One, showing the true purpose of the Dark Ones and how much of a threat they actually are.

This type of emotional manipulation is what makes the player question his previous actions, whether intended or not, it makes the player think wistfully about humanity's role in the world and the meaning behind the Dark Ones' visions.

Sometimes, making the player instantly regret the decision by the realisation of how human perspective isn't always true, but by then it would be too late and the game would end. Making the player realise that the previously thought threat they believed all throughout the game were actually trying to accumulate peace, only attacking on self defence, with the humans, but it was fear that made them seen as hostile.

The alternative ending, is when the player accumulated enough good moral points. Again, this ending also plays with their emotions, because Artyom has deployed a missile beacon, and attacked a dark one acting in self defence by shooting it. The player has a choice in the matter, to just sit back and watch the show, or destroy the missile beacon.

If the player decides to destroy the missile beacon, the dying Dark One at his feet, the one Artyom shot earlier in his vision, mutters "We want peace" and the ending cinematic is played, and it is actually not known whether the dark ones were actually destroyed with Artyom of if they all survived, leaving it on a cliffhanger.

Essentially, this secondary ending has the guidance system falling off the tower, smashing to the ground, and Artyom realizing that the dark ones wanted only to reach out to humanity and make peaceful communications.

All in all, I believe that this is a good method in making the player feel guilty for their actions throughout the entire game, adding onto the emotional setting that the game pulls off very well. This type of player action and play on emotions is the part I personally enjoyed most with the Metro game, it made this game stand out from a wide variety of other games for me.

The other aspect of player actions, is the movement and interaction within the game, which I have explained previously, is very basic with a first person shooter. The player can interact with certain objects in the gameworld such as weapons, corpses for looting, crates also for looting. I have also previously mentioned the interaction with the gasmask and inventory screen. However, since this is a story based game, the interaction the player has with the story itself instead of the gameworld while walking around is considered more important to the developers.

Difficulty and balance

The difficulty of Metro 2033 can sometimes make the game annoying to play, however I have only played the survival mode on hardcore difficulty, so the combat might be slightly easier on lower difficulties, so I will be talking about my experiences in this difficulty and about aspects which easier difficulties don’t actually change.

The biggest problem I had with the game, is how much health the hostiles had, especially the mutated creatures. Some people may say that it is because mutated created have a lot of resistance to bullets due to their super strength. Which is plausible, but I usually find myself wasting 1 or more magazines killing a single enemy, which can be very frustrating since bullets are already hard to find.

Even though humans have a lot less health, and can die with a few shots or sometimes even one, I believe that the payoff still isn't good enough. The reason being is when I scavenge their bodies for supplies, I usually find a small amount of bullets, such as 2 or 3, a filter that lasts a few minutes and sometimes some military grade ammunition.

I find this annoying because the combat of the game isn’t actually that bad, and can be fun against humans, but it’s the fact that supplies are so scarce that the player cannot afford to waste the resources on the enemies, or they will run out during combat.

I believe that this might be less common with a lower difficulty and an easier game mode instead of survival, which focuses on the shooting aspect instead of the survival aspect on the game.

However, a part of the game which I feel isn't very well made and not very balanced is the stealth system of the game. It isn't that the AI isn't very good for stealth, in fact they are very intelligent when it comes to stealth, when compared to other stealth games I played.

It is the fact the stealth in general is very poorly made in my opinion, due to the mechanics not working very well. Especially when the player finds a stealth suit, which makes the player basically invisible in the dark and hides footstep sounds, meaning the player can walk around easily which I find a bit boring and unbalanced.

The most annoying fact about stealth however, is when you miss a headshot with a stealthy weapon such as a throwing knife, all the enemies will suddenly know the player's exact position. When the player is detected, there is no possible way to stop the enemies from pursuing the player, so the player is forced to restart to the last checkpoint or forced to kill every enemy in the area, which brings me back to the fact the ammunition can be wasted.

I believe that the stealth of the game isn't very well made or thought out by the developers, especially with a linear game, where only one route is the actual route, meaning the player cannot create his own.

Which I feel they should have improved with the stealth, since it is fairly often within Metro 2033. Even with a game that has a decent combat system and a very good shooting mechanic, the loss of resources isn't worth it so stealth is a better option, and stealth isn't that well made.

The reason that stealth is such a better option is because of the resources as I have already mentioned, but it cannot be stressed enough how important it is to only use them when needed. The reason is that sometimes the game forces the player into combat situations with humans or mutant creatures, meaning that running out of resources would cause death, forcing the player to play the scene again and running out of resources again. Either creating a deadly loop or forcing the military grade ammunition which is the currency used in the game.

Sometimes, with the forced combat, the game doesn't even go with it's own grain. What I mean by this is that it forces the player to waste a significant amount of ammo on enemies on lengthy gunfights with the only alternative being death.

Overall, with the difficulty and balance of the game, I feel that the developers didn't focus on it enough, because they believed the story and immersion was more important. Which is understandable, but it sometimes disappoints me how they didn't put enough effort of thinking about the stealth and combat of the game.

Feedback

The feedback in Metro 2033 is extremely simple, just like the UI, which I feel can sometimes be a disadvantage. The reason I think this is because there is only one or two ways to find out

about something which can be considered important, some of which may not be obvious to the player.

For example, as I mentioned previously, the filter on the gasmask loses the ability to filter over time, and the only way to tell if by the heavy breathing from Artyom and the bottom of the inventory screen, which doesn't seem very obvious. Which is a bit worrying since this is an important survival aspect of the game, especially in radioactive hot zones and unbreathable air.

Another example is the objectives and the lack of a minimap. I noticed when I click the designated key to open the objectives, it comes up with a lighter on the left hand and a notebook on the right hand, containing a compass with the direction of the objective marked in red and a note with the objectives written.

However, I noticed when I sometimes click the designated button when in the surface or in combat, it only comes up with a lighter since the right hand is occupied with a weapon, which is vital to the situation. Which means that the player will not know where to go when they are forced to carry a weapon, which I sometimes find annoying.

When a level is complete, there is no congratulations screen or mission complete, it just cuts to the loading screen where it shows an image of the map showing the metro system and its stations, and where the player currently is on the said map.

I think that this isn't a bad way of showing a mission complete, because it shows how much the player has progressed, giving a feeling of achievement. Which is what makes some people to carry on playing a bit longer without getting bored off the game.

It is also very simple when the player dies in the game, only showing a screen covered in blood, then cutting to black and automatically reloading the last checkpoint. I find this to be not so bad, but it could definitely could be better, since the player cannot choose what to do with the game until they respawn in the last checkpoint, or that it doesn't give a tip on how to complete the specific part of the game.

Overall, the feedback of Metro isn't very impressive in my opinion, I feel that it could be so much better and that it is a bit too simple to be helpful. I understand that it is doing this in order to be immersive, but the developers sometimes forgot the fun and helpful aspect of games, and that it should be a choice in the options menu to remove parts of the UI for immersion.

Other games such as Tom Clancy's Ghost Recon: Wildlands does the UI customisation perfectly, giving the player the ability to choose what to have in their UI and where it is in the UI, and if the player isn't satisfied he could go back to the default UI layout, or could choose from 3 other presets.

If Metro did their feedback and UI this way, it would of made it much more useful for the game, and if I wanted immersion I would choose it in my will, instead of the game choosing it for me.

Addiction

Personally, I didn't find the Metro 2033 gameplay itself addictive, since it has some frustrating aspects to it sometimes. However, it is the story alone that got me attached to the game, sometimes forcing myself through frustration so I can carry on with the story.

Some people, including the developers, might see this as a good thing because it keeps me playing their game, admiring the work that they have done. However, the way I see it is when players become hooked to the game they become Addicted, as it is very unhealthy can have a negative impact on the mind and body of the player, especially to an extensive amount.

Maybe not from the game itself however, it is from the habit of playing it too much, which can reduce the amount the player will eat, go to the toilet, perform activities such as exercise or even having the lack of sleep.

With Metro, players usually become addicted due to being too immersed in the game as its realism make the gamer feel that they are in the game world, which sometimes happens to myself. Also the fact that the game have unreal features that the gamer may not be able to do in real life such as gunfights, fighting off mutant creatures, explore a post nuclear war world or just found out the desperation of survival.

It is known that video game developers in general, not only Metro, may purposely make the game addictive as they want their game to be the most played video and more popular in comparison to the competition.

This may not be found in Metro, but an example of this in other games is the competition within video games. This makes most the players attached to the game as they believe that playing more will make them better and beat everyone they face. Which is highly addictive to those with a natural competitive mindset, and that it is a fairly common mindset.

Overall, my personal addition in Metro isn't the gameplay, since I don't like it that much, but the story of the Metro game and how I myself have an effect on it.

However, if it weren't for the story of Metro, I would find the game below average as a first person shooter, due to the gameplay aspects not being so great and unpolished, maybe the shooting I find good but everything else tied around it isn't great.

Task 2

(U23:P2) - Explain the fundamental principles of HCI design

Introduction

For this task, I will be talking about the human interactions with video games, using examples from the game I picked for the previous task, which is Metro 2033, included with examples of other video games so I can explain the HCI theory a lot better.

The aims for this task, I will be introducing the elements of the theory on human computer interaction and applying that HCI theory to the game interface design of the game Metro 2033. I will also be explaining the application of the HCI theory on the game control mechanisms.

The key areas of the human computer interaction I will be talking about is the perception of the game, behaviour models of the game, information processing link between the game and the player and specialist interaction to those with a disability or any other form of which has less interaction with hardware.

Perception

Due to the vast amount of personalities in users, the perception of something can be seen differently depending on how the user looks at it. So when it comes to the design of the GUI, the developers had to design it in such a way so that anyone who comes across it can use if well enough without causing any confusion.

When creating the GUI, there may be more than one person involved with the idea, so making sure that everyone has the same understanding of what it will look like.

This is important to carry out otherwise the client of whoever is paying, may not be happy if it's completely different to what he had in mind from the start. Not only that, but they also need to take into account their target audience, and if it's able to be used by disabled users, such as colour blindness or partial blindness.

Colour

It is an obvious fact that colour is an important aspect not only with video games, but with the entirety of human computer interaction. The way coloured are displayed on a monitor, is through the trichromatic system.

Which is a premise where all the colour that are seen from computer display screens give the illusion that there are a lot of colours present, when in fact it is only red, green and blue. So that means, when choosing colours, they have to create a good mix of these three colours to create the colours for the player and for those with colorblindness. So, it's important to ensure that the player has been taken into account, so that it looks attractive and that it doesn't interfere with the gameplay.

It is also important that the developers need to make sure that the colours fit well within the style of the game, and doesn't look obnoxious to the player. Things like this all need to be taken into account when creating a game like Metro 2033, due to its unique style and immersion.

Also for disabled users, such the colour-blind, need to be able to use it. It's no good creating a nice GUI, but the visually impaired cannot use it because things like colours have not been picked correctly for them.

The luminance of the UI in a computer application such as video games is important to make certain objects and other forms of UI stand out to the user, creating more depth for the user. Essentially, this is a system of colours that is used to make sure that the end-user has the best possible visual effects.

The most common type of system being the trichromatic system. Which I have mentioned previously, uses three colour types to display its colours, which are; RED, Green, and Blue (RGB). RGB is the best way to display colours due to the way our eyes work, and how our eyes see these colours on screen.

The best thing for this is to have a certain colour text on a certain background to make sure that the text is easy to read under different conditions. The designers need to take into account, the brightness of user's screens for example.

The pop-out effect is when a selected or active image, such as a button, increases in brightness, to make it more visible and catch the user's attention. This is an effect that is used to make our eyes focus on a specific item or object. They do this by using brighter colours against dull ones.

This technique is used to make sure that the user focuses on a certain element on whatever they are doing, for example, in the inventory screen on Metro 2033 is made in a way that the buttons stand out with the orange colour and the way they pop out, to make sure that the user sees this easily, otherwise they user will have to spend time looking for the information within the inventory screen.

Pattern

The pattern is essentially the template used when creating an interface, making sure that the interface has a consistent pattern to it. Using this allows the user to easily navigate the interface with minor problems. There are certain patterns that game developers look out for, those include: proximity, continuity, symmetry, similarity and connectedness

The proximity is basically the distance between the object that are in an interface. What I mean by this is that the objects have to be far enough so the player can access them with ease and comfortably, without making the user think about the obnoxious distance of the objects.

The continuity is a form of repetition and flow of a specific design. What I mean by this is that the objects are put in a similar pattern so humans can figure out the pattern and be able to move around in digital environments without guessing the next move.

The symmetry is when there is an image or object which is exactly the same on both sides. What I mean by this is that the image would be similar on both sides, instead of looking asymmetric. Which would make certain users annoyed due to the disturbance of the pattern it causes.

The similarity is when the images share similar dimensions or colour, but they have different functions. What I mean by this is when there are different buttons that have the same shape or colour as the rest of the buttons or have a similar consistency with them, but carry out different jobs.

The connectedness is when images are connected by their function. What I mean by this is when a pattern made of buttons in a menu appear as a group, so the user can use common sense to realise they are connected.

An example of a pattern used within a video game is the pause menu in Metro.



As you can see in the screenshot above, all the buttons are always in the same place, easy to read and find, allow the users to navigate easily. It uses the pattern of similarity, continuity, symmetry, continuity and proximity. Meaning that the pause menu in Metro 2033 is a very well designed in the HCI aspect of design.

To elaborate, the pattern on the pause menu is consistent throughout each of the menus, all of which containing text of what the button will access, highlighting the text in a brown colour to show what the user it selecting and also keeping the colours the same so the user wouldn't be confused.

Objects

Objects are essentially the parts that make up the interface. It is used on a UI typically in the form of buttons or sometimes the menus themselves. Typically, a designer will need to take into account the sizes and shapes of the objects, so that it fits in will the background and/or is suitable for the end-user.

The geons are simple 2D shapes such as, cones, circles, rectangles, bricks, etc. that can be easily viewed and given a perspective of 3D. These shapes are quickly recognisable and are normally broken up and arranged, giving the illusion of a 3D object to the human eye.

The gross 3d shapes are 2D shapes that also give the illusion of being 3D shapes to the human eye. To give the illusion, the shape in normally copied and moves slightly higher/lower and to the left/right of the original shape.

Using the example of the pause menu in Metro 2033, the user can tell that the buttons themselves are 2D, but the main menu is in a notepad which has been rendered to look like a 3D object, using gross 3D shapes method to create it.

<u>Behaviour models</u>

Previous successful models or templates are generally used to create a new device or interface as the creators knows that it works, they then just alter it slightly or make it better in some ways.

The reason that they do this is because, it decreases the time needed to think of new designs and decreases resources used as well. Not only that but creating a completely different user interface can make it harder for users to navigate around if they are used to the previous one.

Predictive models

Predictive modelling is the process where the developer tries to predict the outcome of something. For example, when the developer is creating a new game, they have to consider the

features within the game in the design before creating it, such as: what is in the game, the colour scheme, if there is any violence, etc.

Theoretically, the developers can predict the target audience and maybe if it will sell well or not. In relevance to HCI and GUI, the more complex the interface, the smaller the market will be for your product.

With predictive models, there are four main aspects that game developers take in consideration when thinking about predictive models. These four main aspects are the reaction time, Keystroke Level Model (KLM), Throughput (TP) and Fitts' law.

With the reaction time, it is essentially how quickly a display reacts on screen from user input. This is applicable to pretty much everything that the user can interact with ingame, ranging from opening the pause menu to firing a weapon ingame.

The keystroke-level model is used to estimate the time it takes to complete a simple input task using a computer and mouse. By using this method, people normally find more efficient ways to complete a specific task, by looking at what steps are involved in the task, and getting rid of any unnecessary steps or rearranging the steps to make it more efficient.

The throughput This is used to estimate how long it will take to transfer a specific amount of data from location to another and/or the amount of work the computer can do/handle in a given time period. An example of this would be how long it takes to transfer data from your computer to a USB memory stick.

Fitts law Fitts' law is an equation that is used to calculate the time it takes for you to reach a target object. For example, the time it takes for you to move your mouse to click on a button or link on a website. The equation looks like this: $T=a+b \log2 (2 d/w)$.

'T' stands for time, which is the amount of time it takes for you to get to your targeted object 'a + b' stands for the two coefficients. A is the start/stop time of the device. B is the speed of the device. D stands for distance to the targeted object. W stands for the width to the targeted object (on an axis).

Game developers use this aspects on behaviour models to make the game as responsive as possible. While playing Metro, I haven't had any issues with interacting with the game and never experience any form of game lag, meaning that the game is well optimised and the predictive models were very well made in the game.

Descriptive models

Key action model, or KAM, is a variety of different functions that each key mapped inputs have. This type of descriptive model is determined by the symbol found on the key. For example, the

keyboard on a computer game would usually have the W, A, S and D keys for movement and mouse 1 button to fire a weapon.

The buxton's three state model is where the devices must be ergonomic to the user and have three main states, which are responsive, easy to use and adjustable. To elaborate, three state model focuses on the user moving and using the mouse, touch pad, or the mouse wheel.

The first state, which is responsive, is the human contact with the mouse. The second state, which is easy to use, is moving the tracking symbol (the cursor). The third state, which is adjustable, is moving something on the desktop, for example an icon.

Guiard's model states that not all the input methods are really suitable for everyone. For instance, there are people who are both left and right handed, and there are also users who have disabilities and therefore need different input methods to fit their needs.

Basically, when a developed is creating a game, they need to focus on this area in order to fit a wider audience. They have to make sure that the game is created in such a way that both left-handed and right-handed people can use the effectively. For example, a keyboard or a mouse, so the user will have the option to change what buttons do what.

Information Processing

This section is more about what the humans do to deal with the information we are given and what we do with it after. Essentially, this is where we get both the human way of processing information and the way the computer does it, and game developers use this to build better computer interaction.

Basically, this allows the developers to build a more efficient way of communicating with each other. Although, without this, it would be incredibly hard to use new products because the user would have to learn a new interface every time they buy a new product.

Humans

The human as a component basically means that by treating humans as a component. Basically, this is so the developers can make interfaces that are designed around us and are perfect for humans as a whole. This means that the end-user will have mostly what it needs.

When dealing with HCI, it is important for the designers to think about who will be using it, which is the target audience in general. This way, the developers can create it around that. Usually, the designer will have to think about what the requirements of the end-users are, so the designers would need research about this specific topic a lot.

The human information processing is what the designers and developers of a game need to make sure that the interface is easy to understand and optimal for the end-user, which comes under HIP, or Human Information Processing. This principle is used to analyse how we, as humans, process certain types of information and how we then input it into a HCI machine of some type.

Many people believe that humans and computers both have similar way of storing data, long-term and short-term. However, I highly doubt this since computers are extremely fast when handling information such as binary, which isn't even thinkable to do as a human.

Computers on the other hand, are able to perform calculations much faster than we can, and also go on for much longer without slacking, so in some respect computers are more efficient, but, don't forget that it's humans that input the data in the first place, as computers cannot think for themselves.

The goals, operators, methods and selection, also known as GOMS, is concerned with how the goals, or the outcomes, can be achieved by operating, or interacting, with different methods, whether it is more than one way to carry out a specific task, to select from. To elaborate, i will be explaining the goals, operators, methods, and selection rules.

With the goals, these are what people are aiming towards by the end. For instance, when working on a task, the player will have short-term goals and long-term goals that the user and/or your team will be working towards, such as the end of a game or the end of a mission.

With the operators, these basically are the actions that need to be taken in order for the intended task to get done. For example, in order to go onto another mission within the game, the user will need to move through the checkpoints or objective, and complete the tasks in order to reach the end of the mission.

With the methods, this is the plan that is going to be used in order to get something done. For example, if the user wanted to move to a checkpoint within the game, the user will first have to think about using the movement keys to move within the game, the user will then need to move towards the checkpoint by using the UI or any other form of information to figure out where it is.

With the selection rules, this is the method the user will use in order to complete a specific task. First of all the user will need to identify what the task that the user is going to do, then the user must think about what the most efficient method is to complete the task.

Overall, these aspects are important when creating a video game because the human interaction is what makes the game carry out its purpose. If it does not have a good information processing towards humans, then the game would not work very well as it is intended to, beating the entire purpose of a video game.

Specialist

The specialist aspect of HCI is the concern with the accessibility for unique or special features or scenarios, such as user with disabilities that aren't able to interact with the hardware or software which was made ergonomically to users who don't have a form of physical disability.

For example, a form of specialist HCI would be the software and hardware specifically created for those found with significant physical disabilities, such as physically restricting disabilities such as arthritis or the lack of limbs, meaning that they need to use a completely different body part.

The type of hardware can range from touch screens, mouse and keyboard alternatives and other PC computer alternatives such as notetakers. The type of software can range from an on-screen keyboard software to a software for alternative computer input. Which make the access of a PC usable by those affected with a physical disability.

However, it isn't always physical disabilities, it can also be disabilities with memory or the mind in general. Software such as a beginner's guide to computers, laptops and tablets to reminds those with impaired memory to learn how to use a computer again.

From my experience, I haven't found many games with a vast specialist human computer interaction, but I have noticed some small addition to games such as a colourblind function and the remapping of keys on a keyboard.

For instance, with Metro 2033, I haven't noticed many forms of specialist human computer interactions. The only method I have seen through the options menu is the ability to remap the keys on the keyboard, so the game doesn't even have a color blind option.

Which is a disadvantage due to the fact the game doesn't fit well with those with disabilities such as colour blindness, but it can be used by both left and right handed people. So the audience isn't reduced as significantly, however the game developers should have considered the colour blind players.

However, I reckon it might be the fact that most the specialist HCI is mostly hardware. So it would make sense if the game didn't have it inbuilt into the options menu, but instead includes support for these type of devices. Although, I still believe that Metro 2033 should have implemented a colourblind mode in the options.

Conclusion

In conclusion, the way I see it, Metro 2033 was mostly made for immersion and storytelling for the soul purpose to manipulate the players emotions, which I can agree was extremely well made and it was the reason I kept playing the game.

However, the reason the game had such a good, immersive story and visuals is the same reason the gameplay is mediocre at best. I personally didn't enjoy the gameplay that much, but it had its fun moments, especially with the shooting.

Honestly, if the game didn't have such an immersive and enjoyable story, I probably wouldn't of played the game that much, since there are other games I play with much better gameplay mechanics.

Although, I would still recommend getting Metro 2033, especially the redux edition, because of the beautifully remastered graphics which makes it so much more immersive. Also, I would recommend getting its sequel, Metro Last Light, which has an equally good story as the first one, some would even consider a better story than the first, which is rare to find in video games.

From my personal recommendation, for the most immersive and realistic experience, I would recommend playing with the survival gamemode and hardcore difficulty, if you are used to first person shooters, or regular difficulty, if you are not so used to first person shooters. I would also recommend turning the crosshair off, making the spoken language Russian with english subtitles, so the setting of the game feels a lot more genuine and immersive.

Overall, I would recommend Metro 2033 and leave it for the player to judge whether the story is worth going through over the gameplay, which isn't horrible but it isn't amazing either. I personally think that the story compensated well with the mediocre and sometimes unbalanced gameplay mechanics.

However, if the player gets easily impatient, frustrated or scared I would be careful playing Metro 2033 because it likes to manipulate this emotions, whether intentional or not. Also, if the player gets easily emotional, it would make the moral of the story much more impacting, meaning that the player would reflect on the story a lot more.